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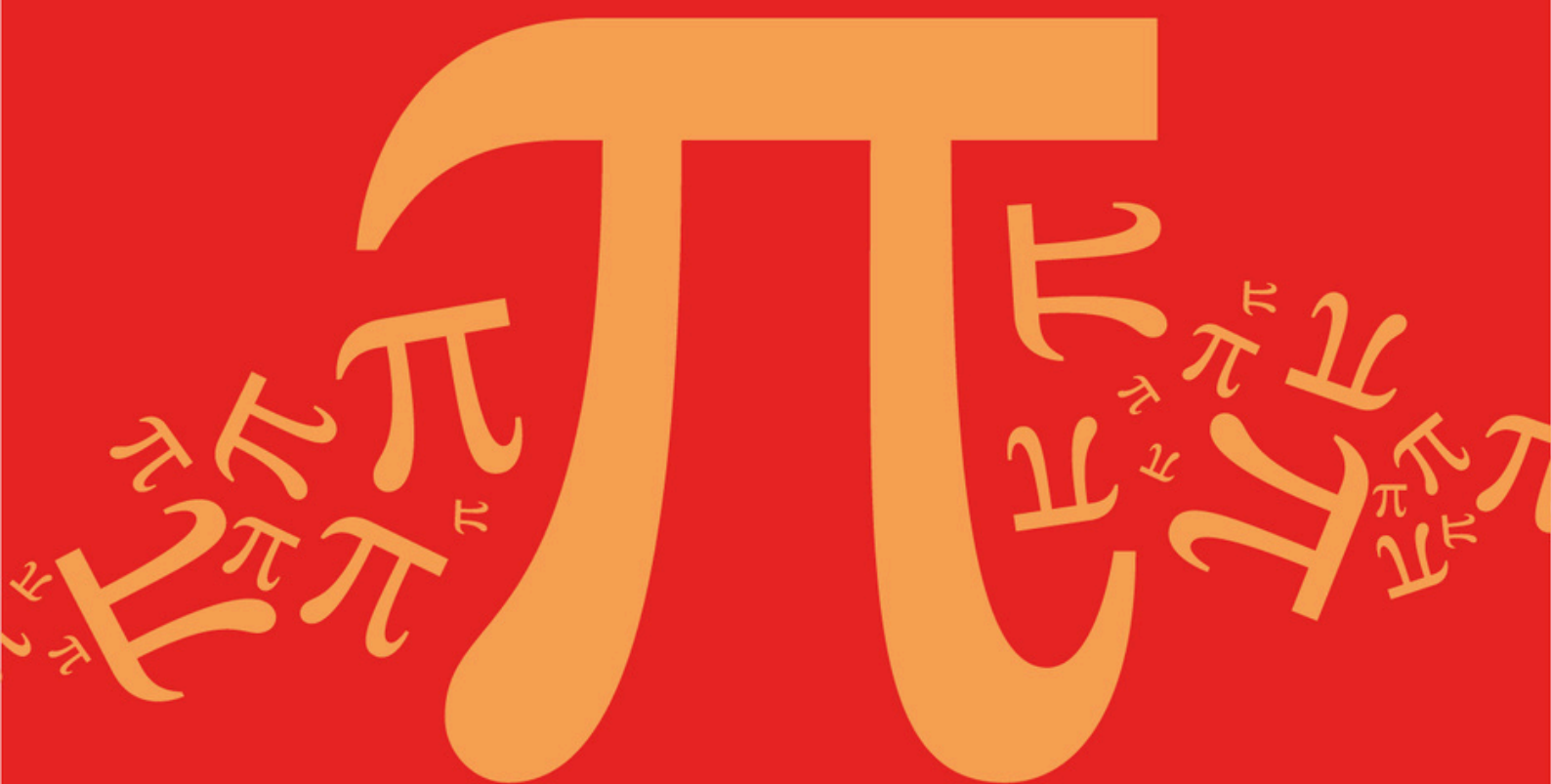
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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D16: Transformations of Graphs – Rotations and Enlargements

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

D: Further Algebra and Functions

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D16: Transformations of Graphs – Rotations and Enlargements

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

The plane transformation T is a rotation through θ radians anticlockwise about O , and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $c = \cos \theta$ and $s = \sin \theta$.

- (a) Write down the inverse of the matrix $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and hence show that

$$x = cX + sY \quad \text{and} \quad y = -sX + cY$$

(3)

- (b) The curve C has equation $x^2 - 6xy - 7y^2 = 8$.

The image of C under T is the curve C' with equation $pX^2 + qXY + rY^2 = 8$.

- (i) Use the results of part (a) to show that

$$q = 6s^2 + 16sc - 6c^2$$

and express p and r similarly in terms of c and s .

(4)

- (ii) Given that θ is an acute angle, find the values of c and s for which $q = 0$ and hence in this case express the equation of C' in the form

$$\frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1$$

(8)

- (iii) Hence explain why C is a hyperbola.

(1)

(Total 16 marks)